

**2025**



# Carbon Emissions Report 2025





### Introduction

Kumaraguru Institutions is a multi-campus educational institution in Tamil Nadu, South India, offering undergraduate, postgraduate, and research programs in Engineering, Technology, Management, Agriculture, Liberal Arts, and Science. Guided by its motto, "Character is Life," the institution is committed to academic excellence, holistic development, and nurturing responsible individuals with integrity and purpose. Through experiential learning, innovative research, and strong community engagement, Kumaraguru fosters a culture of inquiry and social responsibility, empowering students to address real-world challenges and contribute meaningfully to society.



Microcosm is a long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance, resource efficiency, and responsible environmental stewardship.

This document follows Greenhouse Protocol for GHG emissions and provides a

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### Microcosm Logo

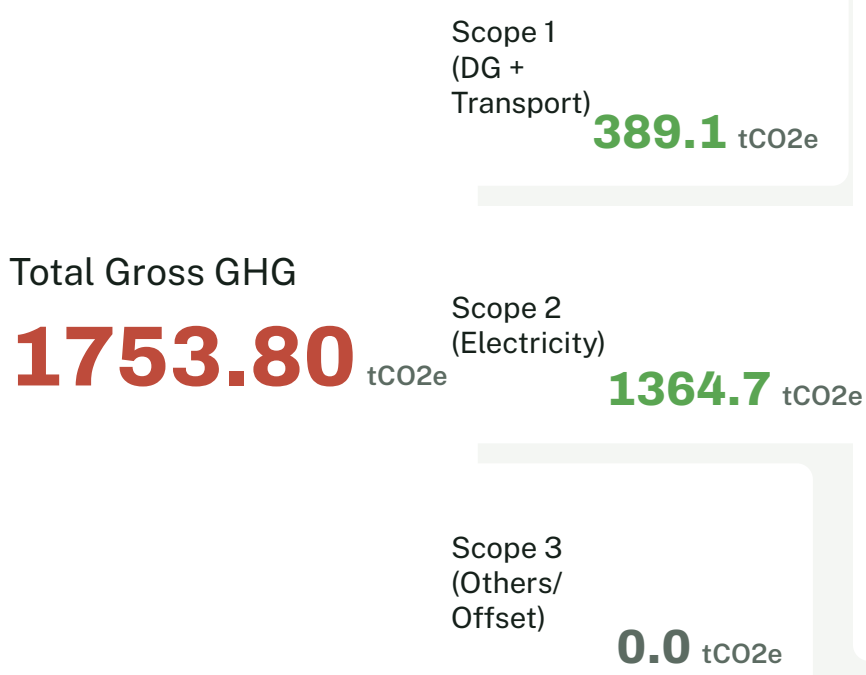
Microcosm is a long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance, resource efficiency, and responsible stewardship.

This Carbon Emissions Report provides a transparent account of our Greenhouse Gas (GHG) inventory across energy, transport, and waste. By quantifying these operational impacts, we establish a robust baseline for targeted reduction strategies and reaffirm our continuous commitment to environmental sustainability.

# Highlights

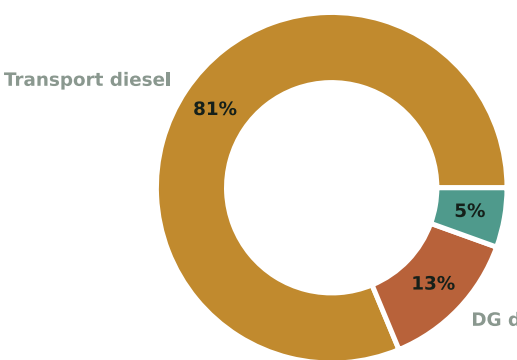


## EMISSION PROFILE

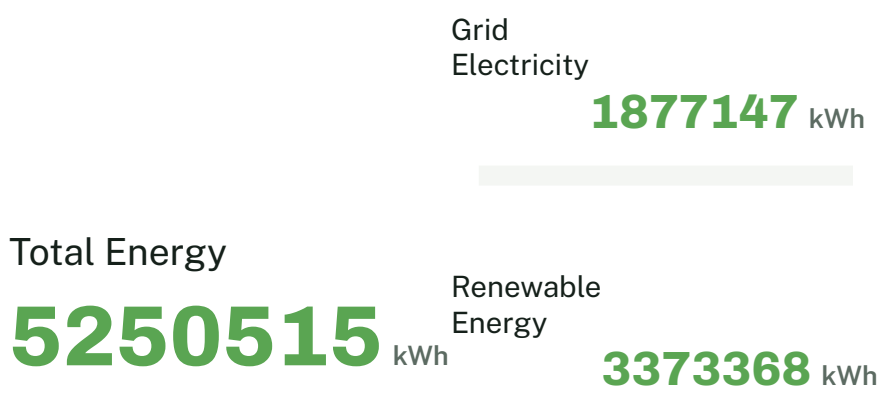


### Emission Contribution

#### Scope 1 Breakdown

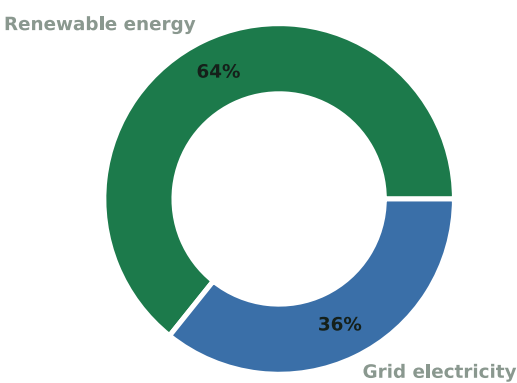


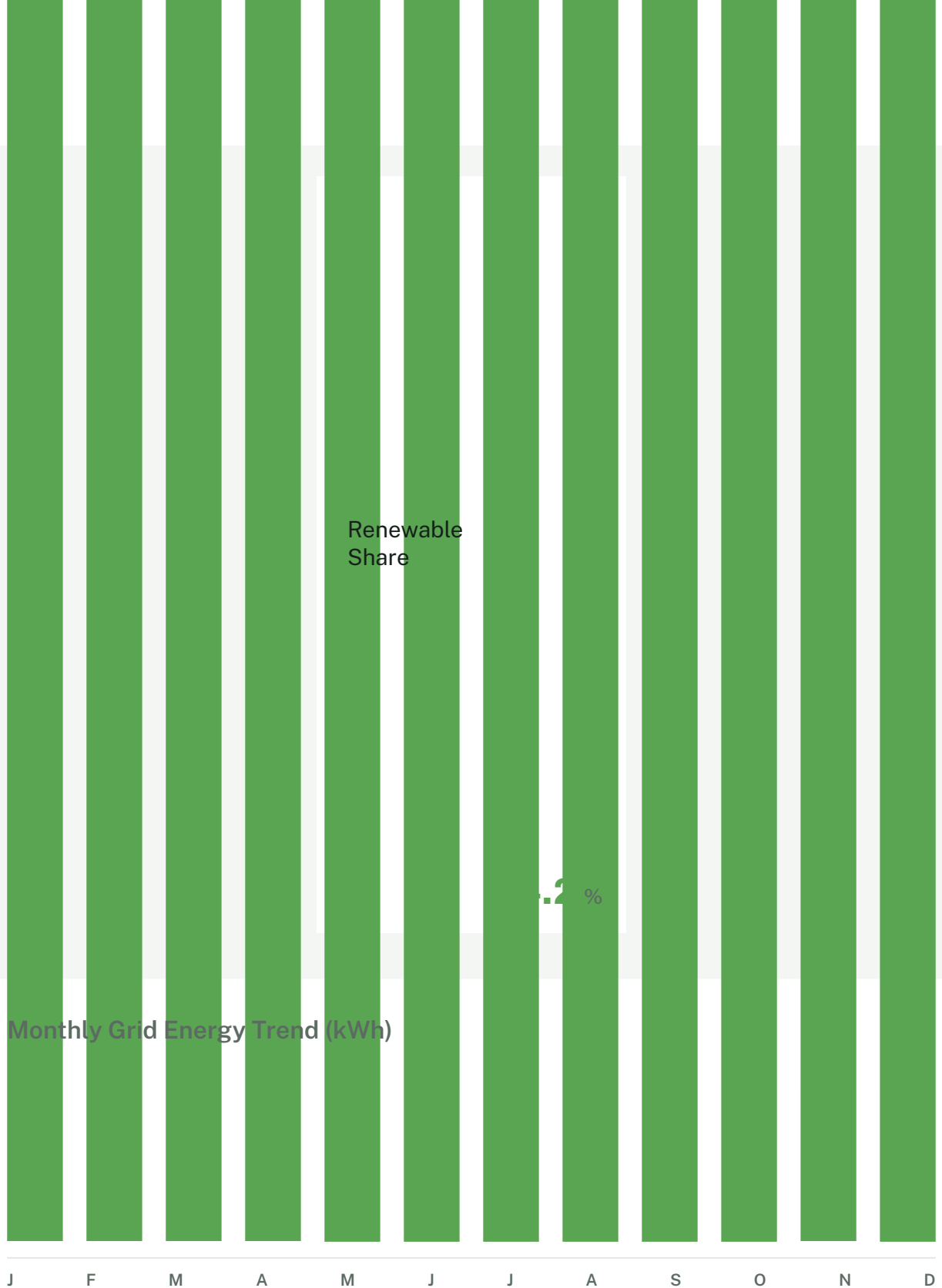
## ENERGY INDICATORS



### Energy Split

#### Renewable Mix





# Transport Indicators



## FLEET & FUEL OVERVIEW

Total Fleet Fuel Consumption

**126034** Litres

Fleet Carbon Footprint

**337.6** tCO<sub>2</sub>e

Petrol vs Diesel (Litres)

**8954.1** L

Petrol (7.1%)

**117080.0** L

Diesel (92.9%)



Monthly Transport Fuel  
(Petrol+Diesel)

Petrol vs Diesel (Emissions)

**21.4** tCO<sub>2</sub>e

Petrol (6.3%)

**316.2** tCO<sub>2</sub>e

Diesel (93.7%)



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### Transport Fleet Insights

Transport emissions decreased by 0%. Continued fleet optimization is contributing to this reduction.

# Climate Action & Mitigation



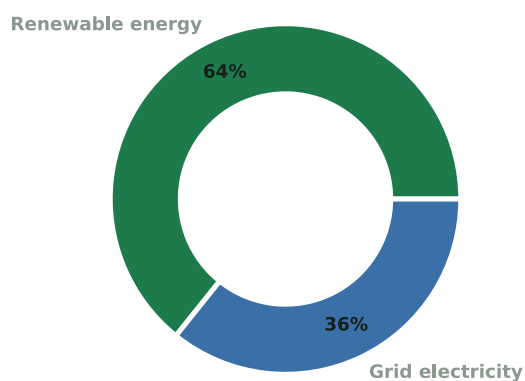
## SUSTAINABILITY INITIATIVES

Total Avoided Emissions via Renewables

**2452.4** tCO<sub>2</sub>e

### Renewable Integration Impact

#### Renewable Mix



### Per Capita Intensity

**0.000** tCO<sub>2</sub>e / person

### Net Zero Progress Target (Illustrative)

|                                  |                                  |
|----------------------------------|----------------------------------|
| <b>2452.4</b> tCO <sub>2</sub> e | <b>1753.8</b> tCO <sub>2</sub> e |
| Avoided (58.3%)                  | Remaining (41.7%)                |



### Decarbonization Story

Renewable energy integration contributed 3373368.00 kWh, offsetting approximately 2452.44 tCO<sub>2</sub>e of potential emissions. Our current renewable share stands at 64.2%.

# Executive Summary

High-level overview of carbon performance for Carbon Emissions Report - 2025.

## Performance Overview

During this reporting period, the campus generated 1753.80 tCO<sub>2</sub>e of gross carbon emissions. Through our renewable energy initiatives, we successfully avoided 2452.44 tCO<sub>2</sub>e, resulting in a net carbon impact of -698.64 tCO<sub>2</sub>e. We remain committed to our long-term decarbonization pathway.



## CARBON STORY

Gross GHG Emissions

**1753.8** tCO<sub>2</sub>e

Net GHG Emissions

**-698.6** tCO<sub>2</sub>e

Avoided Emissions

**2452.4** tCO<sub>2</sub>e



## STRATEGIC RECOMMENDATIONS

- Accelerate the transition of internal campus mobility to Electric Vehicles (EVs) to reduce Scope 1 transport emissions.
- Expand rooftop solar capacity to push the renewable energy mix beyond the current baseline.
- Implement automated BMS (Building Management Systems) for high-load commercial blocks to optimize HVAC consumption.

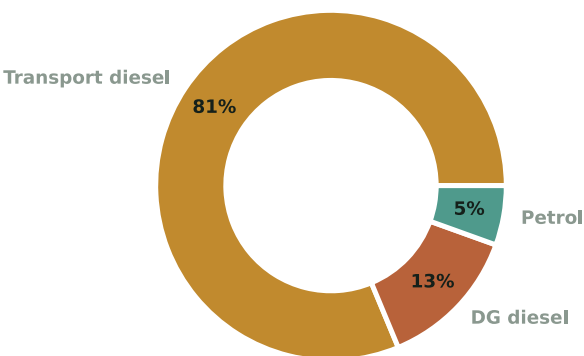
# Scope 1 Story

## Direct Emissions Overview

Scope 1 emissions generated 389.11 tCO<sub>2</sub>e. Transport remains the dominant emission source in this category. Overall Scope 1 emissions decreased by 0% compared to the previous period.

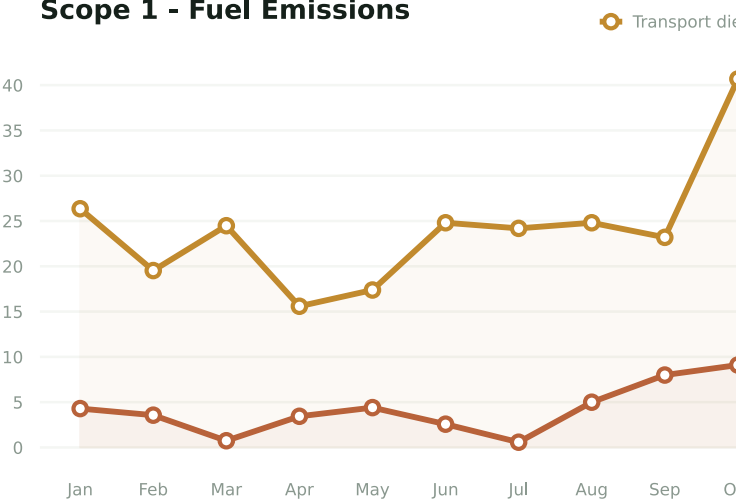
### Scope 1 Composition

#### Scope 1 Breakdown



### Emissions Trend

#### Scope 1 - Fuel Emissions



## Transport Deep Dive

Transport emissions decreased by 0%. Continued fleet optimization is contributing to this reduction.

# Transport Emissions (Scope 1)

**337.6** tCO<sub>2</sub>e

| MONTH | PETROL<br>(LITRES) | EMISSIONS<br>(TCO <sub>2</sub> E) | DIESEL<br>(LITRES) | EMISSIONS<br>(TCO <sub>2</sub> E) |
|-------|--------------------|-----------------------------------|--------------------|-----------------------------------|
| JAN   | 746.0              | 1.7814                            | 9757               | 26.3537                           |
| FEB   | 483.5              | 1.1546                            | 7230               | 19.5282                           |
| MAR   | 493.0              | 1.1773                            | 9063               | 24.4792                           |
| APR   | 252.0              | 0.6018                            | 5771               | 15.5875                           |
| MAY   | 440.0              | 1.0507                            | 6435               | 17.3809                           |
| JUN   | 414.0              | 0.9886                            | 9184               | 24.8060                           |
| JUL   | 853.0              | 2.0370                            | 8955               | 24.1875                           |
| AUG   | 908.0              | 2.1683                            | 9184               | 24.8060                           |
| SEP   | 1123.0             | 2.6817                            | 8588               | 23.1962                           |
| OCT   | 976.2              | 2.3312                            | 15060              | 40.6771                           |
| NOV   | 1124.6             | 2.6855                            | 13629              | 36.8119                           |
| DEC   | 1140.8             | 2.7242                            | 14224              | 38.4190                           |

## Transport Insights

The table above highlights the monthly breakdown of combustion fuels used in institutional mobility. Diesel remains the primary fuel source for heavy transport and buses.

Cumulative Fuel Mix

8954.1 L  
Petrol (7.1%)

117080.0 L  
Diesel (92.9%)



# DG Set Emissions (Scope 1)

**51.5** tCO<sub>2</sub>e

| MONTH | DIESEL (LITRES) | EMISSIONS (TCO <sub>2</sub> E) |
|-------|-----------------|--------------------------------|
| JAN   | 1588.62         | 4.2909                         |
| FEB   | 1319.34         | 3.5635                         |
| MAR   | 274.23          | 0.7407                         |
| APR   | 1273.80         | 3.4405                         |
| MAY   | 1628.22         | 4.3978                         |
| JUN   | 951.39          | 2.5697                         |
| JUL   | 214.17          | 0.5785                         |
| AUG   | 1849.32         | 4.9950                         |
| SEP   | 2959.11         | 7.9926                         |
| OCT   | 3368.64         | 9.0987                         |
| NOV   | 2162.82         | 5.8418                         |
| DEC   | 1475.43         | 3.9851                         |

## DG Operations Analysis

Diesel Generator usage is strictly reserved for grid outages and critical power backups. Optimizing backup load management helps minimize these localized emissions.

### DG Monthly Trend

J F M A M J J A S O N D

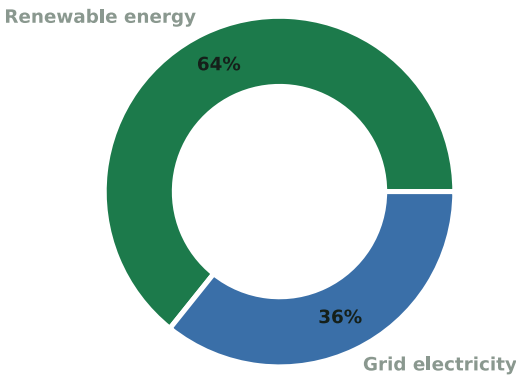
# Scope 2 Story

## Indirect Emissions Overview

Grid electricity consumption accounted for 1364.69 tCO<sub>2</sub>e. This represents 77.8% of total gross emissions. Scope 2 emissions fell by 0%. Continuous efforts in optimizing HVAC and lighting can further reduce this footprint.

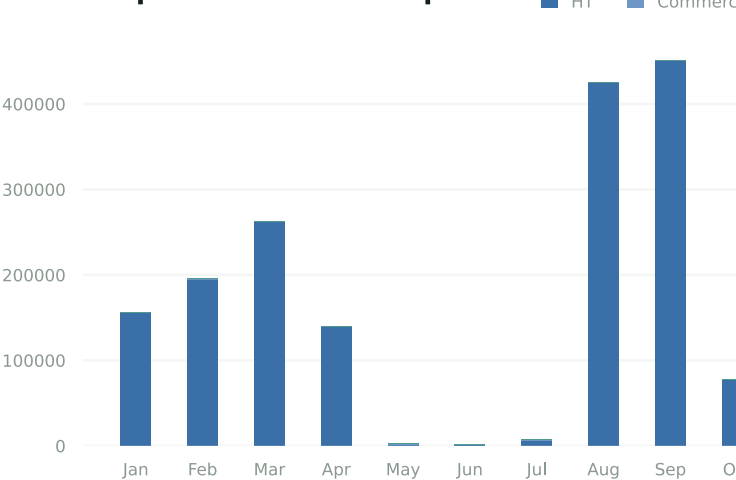
### Renewable Energy Mix

Renewable Mix



### Grid Consumption Profile

Scope 2 - Grid Consumption



### Renewable Impact

Renewable energy integration contributed 3373368.00 kWh, offsetting approximately 2452.44 tCO<sub>2</sub>e of potential emissions. Our current renewable share stands at 64.2%.

# Electricity Emissions (Scope 2)

1364.7 tCO2e

RENEWABLE ENERGY

3373368 kWh

GRID ENERGY

1877147 kWh

TOTAL CONSUMPTION

5250515 kWh

| MONTH | GRID HT              |                      | GRID COMMERCIAL      |                      | GRID TEMPORARY       |                      |
|-------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|       | ELECTRICITY<br>(KWH) | EMISSIONS<br>(TCO2E) | ELECTRICITY<br>(KWH) | EMISSIONS<br>(TCO2E) | ELECTRICITY<br>(KWH) | EMISSIONS<br>(TCO2E) |
| JAN   | 155222               | 112.846              | 1023                 | 0.744                | 0                    | 0.000                |
| FEB   | 194516               | 141.413              | 862                  | 0.627                | 0                    | 0.000                |
| MAR   | 261397               | 190.036              | 1119                 | 0.814                | 0                    | 0.000                |
| APR   | 139071               | 101.105              | 1140                 | 0.829                | 0                    | 0.000                |
| MAY   | 1321                 | 0.960                | 1001                 | 0.728                | 0                    | 0.000                |
| JUN   | 690                  | 0.502                | 1023                 | 0.744                | 0                    | 0.000                |
| JUL   | 5934                 | 4.314                | 1128                 | 0.820                | 428                  | 0.311                |
| AUG   | 424410               | 308.546              | 1094                 | 0.795                | 416                  | 0.302                |
| SEP   | 450216               | 327.307              | 1027                 | 0.747                | 389                  | 0.283                |
| OCT   | 76958                | 55.948               | 1000                 | 0.727                | 346                  | 0.252                |

| MONTH | GRID HT              |                      | GRID COMMERCIAL      |                      | GRID TEMPORARY       |                      |
|-------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|       | ELECTRICITY<br>(KWH) | EMISSIONS<br>(TCO2E) | ELECTRICITY<br>(KWH) | EMISSIONS<br>(TCO2E) | ELECTRICITY<br>(KWH) | EMISSIONS<br>(TCO2E) |
| NOV   | 105958               | 77.031               | 963                  | 0.700                | 331                  | 0.241                |
| DEC   | 46972                | 34.149               | 899                  | 0.654                | 293                  | 0.213                |

### Grid Operations Insights

The detailed monthly tracking indicates peak grid consumption variations. Industrial High Tension (HT) lines remain the largest contributor to our Scope 2 profile.

#### Grid vs Renewable Generation

**1877147.0** kWh  
Grid (35.8%)

**3373368.0** kWh  
Renewable (64.2%)



# Emissions Comparison

Performance analysis vs previous reporting period



## CONSOLIDATED GHG SUMMARY

| CATEGORY               | CURRENT PERIOD | PREVIOUS PERIOD | % CHANGE |
|------------------------|----------------|-----------------|----------|
| Total Emissions        | 1753.80        | N/A             | 0.00%    |
| Scope 1 Emissions      | 389.11         | N/A             | 0.00%    |
| Scope 2 Emissions      | 1364.69        | N/A             | 0.00%    |
| Renewable Energy (kWh) | 3373368        | N/A             | 0.00%    |

### Current vs Previous Gross Emissions

1753.8 tCO<sub>2</sub>e  
Current (50.0%)

1753.8 tCO<sub>2</sub>e  
Previous (50.0%)



### Progress Summary

The comparison table above illustrates our performance trajectory against the previous baseline. Reductions in total emissions are primarily driven by continuous optimization in energy usage and sustained renewable integration.

# Methodology & Conclusion

## Reporting Standards and Final Insights

### Purpose of the Report

This report presents the Greenhouse Gas (GHG) emissions inventory for Kumaraguru Institutions for the reporting period, compiled in accordance with ISO 14064-1:2018. The objective of this inventory is to measure and disclose the institution's GHG emissions, and provide reliable data to support future emissions reduction strategies.

### GHG Quantification Approach

The inventory has been developed using an activity-based quantification approach. Emissions were calculated by multiplying activity data (e.g., fuel consumption, electricity consumption) with appropriate emission factors derived from recognized national and international sources (IPCC, CEA, DEFRA).

### Organizational Boundary

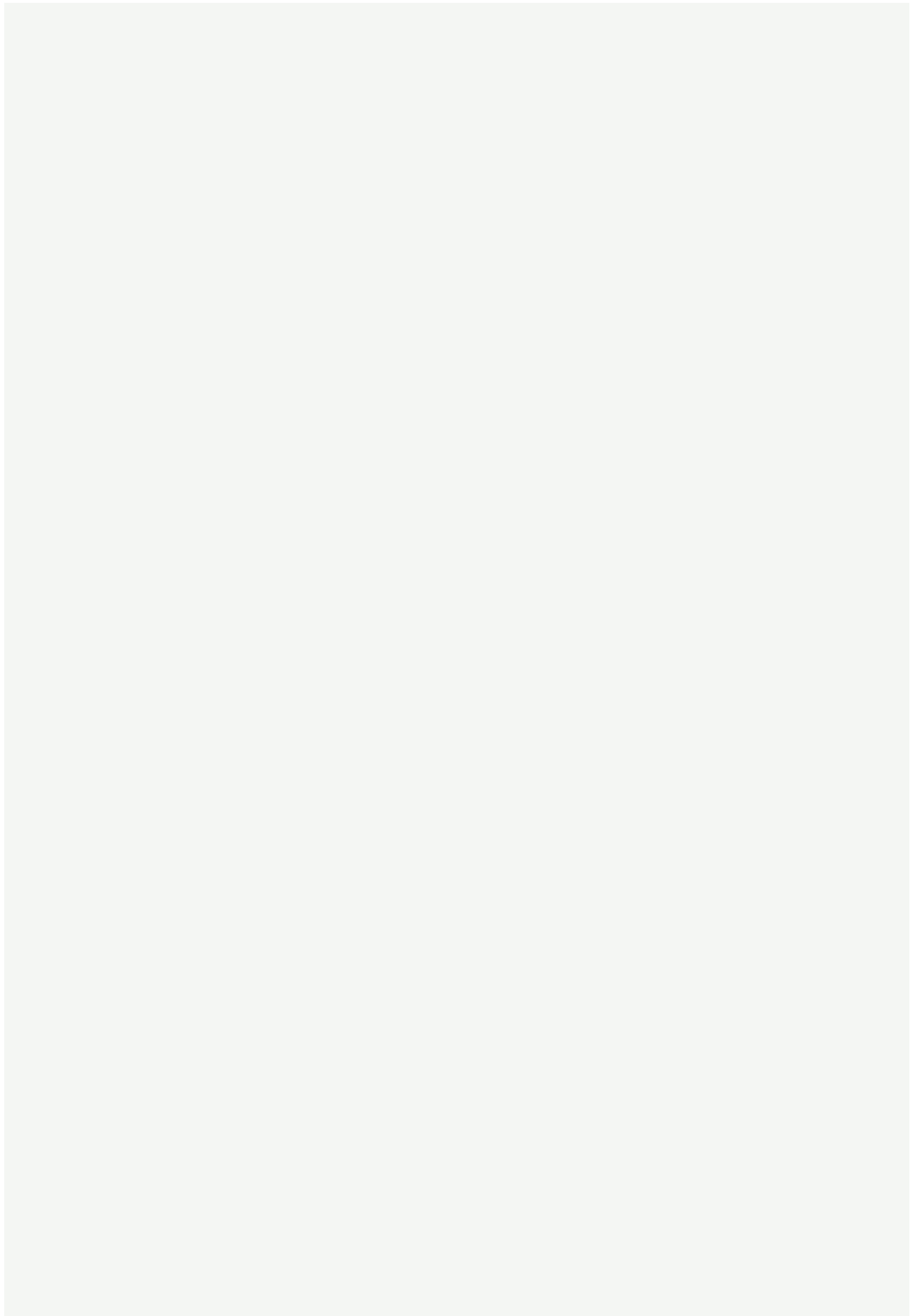
The organizational boundary has been defined using the operational control approach. All facilities and operations over which Kumaraguru Institutions has operational control are included. Scope 3 emissions have been assessed at a screening level but are excluded from quantitative reporting due to the unavailability of consolidated data.

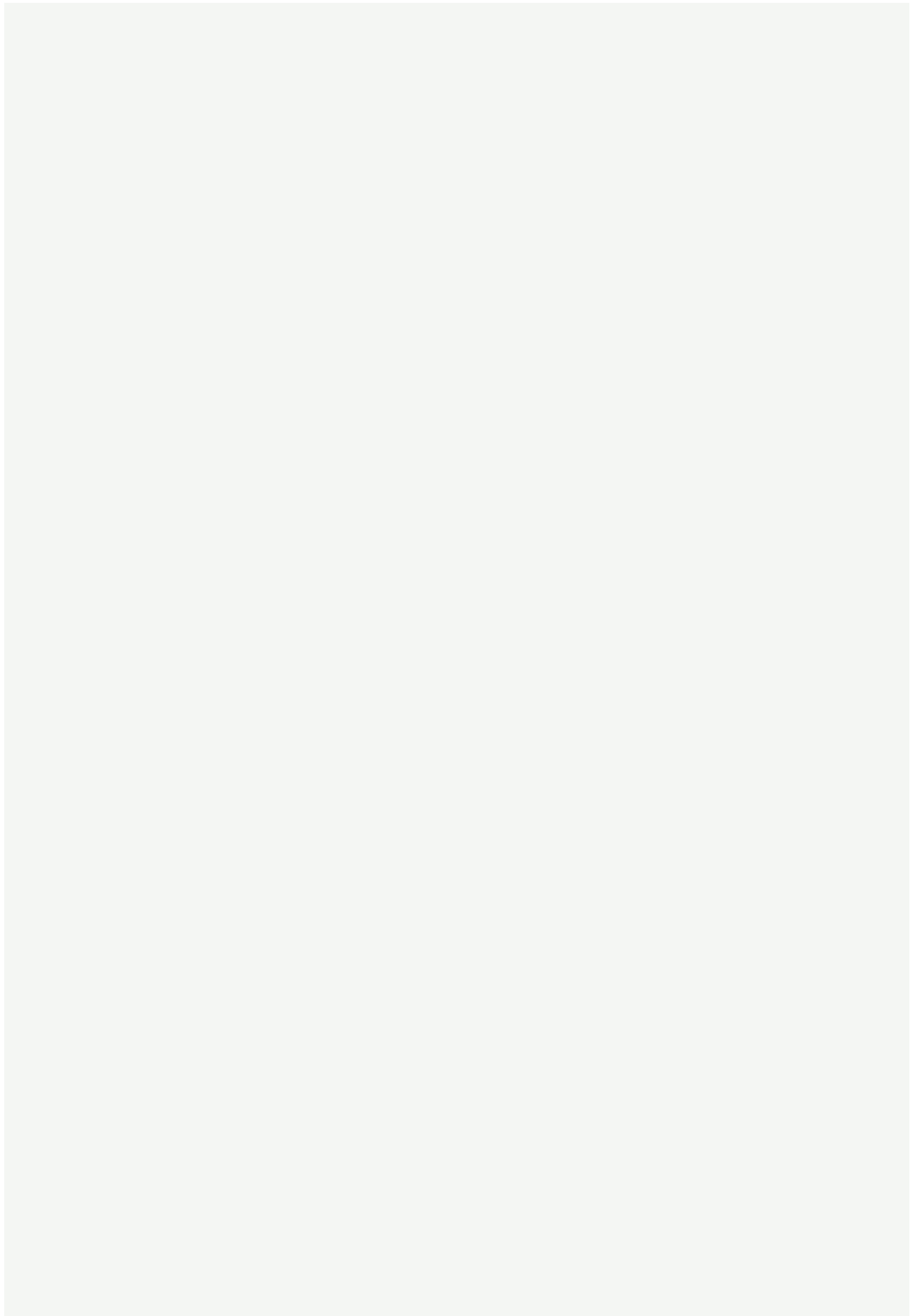
### Standard Emission Factors Applied

| EMISSION SOURCE       | FACTOR | UNIT                      |
|-----------------------|--------|---------------------------|
| Grid Electricity (TN) | 0.727  | kgCO <sub>2</sub> e / kWh |
| Diesel Fuel           | 2.701  | kgCO <sub>2</sub> e / L   |
| Petrol Fuel           | 2.388  | kgCO <sub>2</sub> e / L   |

### Conclusion

This GHG inventory enhances the understanding of emission sources within campus operations and provides a foundation for tracking environmental performance. The institution intends to progressively improve data quality, expand emission coverage, and integrate climate considerations into operational decision-making.





REPORT PREPARED BY

## **Microcosm Sustainability Intelligence**

A long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance.